Cumulative Meta-Analysis

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Introduction

Cumulative meta-analysis plots the running pooled estimate as studies are added one by one in chronological (or any pre-specified) order. Each successive pool incorporates one more study, producing a sequence of estimates that traces how the body of evidence evolved over time. The resulting forest plot reveals when the cumulative effect first crossed a meaningful threshold — clinical relevance, statistical significance, or stability — and whether later studies overturned, attenuated, or simply added precision to the early consensus. Cumulative meta-analysis is widely used in retrospective evidence syntheses to assess research efficiency: if the evidence had been pooled in real time, when would the answer have been clear, and how many subsequent confirmatory trials were arguably redundant?

Prerequisites

A working understanding of inverse-variance meta-analytic pooling, the sequential-analysis idea, and an awareness of why repeated looks at accumulating evidence raise multiplicity concerns when the goal is decision-making rather than description.

Theory

For an ordering of studies $1, \ldots, k$, the cumulative estimate at step t is the pooled effect computed from studies $1, \ldots, t$ under the chosen meta-analysis model. Plotting these $\hat{\theta}_{(1:t)}$ with their confidence intervals along the ordering axis (typically year) reveals stability and convergence: a horizontal band of overlapping CIs indicates the answer settled early, while a shifting estimate signals that later evidence materially changed conclusions.

The same procedure can use precision-based ordering (largest studies first) to assess sensitivity to the small-study effect, or risk-of-bias ordering (low-risk studies first) to assess whether pooled conclusions hinge on lower-quality evidence.

Assumptions

Study ordering is correctly recorded; studies are independent; the meta-analysis model used at each step is the same as for the full pool. Cumulative meta-analysis is descriptive, not inferential — formal sequential-testing methods such as trial-sequential analysis adjust for repeated looks; ordinary cumulative meta-analysis does not.

R Implementation

```r
library(metafor)

set.seed(2026)
k <- 12
year <- sort(sample(1990:2023, k))
yi <- rnorm(k, 0.35, 0.2)
vi <- 1 / sample(50:250, k, replace = TRUE)
```

```r
df <- data.frame(year, yi, vi)

res <- rma(yi, vi, data = df, method = "REML")
cum <- cumul(res, order = df$year)

plot(cum, main = "Cumulative meta-analysis by year",
     xlab = "Year")
```

Output & Results

`cumul()` returns a running-pooled estimate, confidence interval, and heterogeneity statistics at each step in the specified ordering. The associated `plot()` method renders a forest-style display where each row corresponds to "evidence after the t-th study", making convergence visually obvious. Many pooled effects stabilise after four to six well-powered studies; remaining studies typically tighten the CI without shifting the centre.

Interpretation

A reporting sentence: "Cumulative meta-analysis ordered by publication year showed that the pooled effect became statistically significant after the third study (1994) and remained essentially unchanged through 2023; the 18 trials published thereafter added precision but did not alter the direction or magnitude of the conclusion, suggesting an opportunity for earlier evidence-based decision-making had a living synthesis been in place." Always describe both when the evidence stabilised and what later studies contributed.

Practical Tips

- Order primarily by publication date for the temporal interpretation; precision-based and risk-of-bias orderings are useful complementary sensitivity analyses.
- Cumulative meta-analysis is descriptive — for inferential stopping rules under repeated looks, use trial-sequential analysis (TSA) with appropriate alpha-spending.
- Convergence to stability after the first few studies often reveals research waste: many subsequent trials confirmed what was already known.
- Combine with meta-regression on year to formally test whether effect sizes drifted with time, e.g. as practice patterns or populations changed.
- Plot both point estimate and confidence interval; a shrinking CI without estimate movement is a different signal from a moving estimate.
- Use cumulative meta-analysis to motivate living-systematic-review infrastructure: continuous evidence updating prevents the "many years late" problem that retrospective cumulative plots so often expose.

R Packages Used

`metafor::cumul()` for the canonical cumulative pooling and its `plot()` method; `meta::metacum()` for an alternative interface aligned with the `meta` package's forest-plot conventions; `RTSA` and `gsDesign` for trial-sequential analysis when inferential adjustment for multiple looks is required; `metaviz` for richer cumulative-forest visualisations.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis